

Regression methods $x \in \mathbb{R}^p$, $y \in \mathbb{R}$ n = sample size
 (low bias, high var.) $\swarrow$ careful with overfitting

Method	Model para.	Tuning para	High complexity, if...	Fitting
k NN	—	$k = 1, \dots, n$	small k	—
Linear reg.	$\vec{\beta} = (\beta_0, \dots, \beta_p)$ $\in \mathbb{R}^{p+1}$	—	(p large)	LS closed form / GD
Lasso/ridge	— " —	penalty $\lambda \geq 0$ $\lambda < 0 \Rightarrow LM$	small λ	GD (lasso) closed form (ridge) penalized LS
Reg. trees	(splitting vars splitting points)	nb. of leaves $ T $ or: cost complexity para $\alpha \geq 0$	$ T $ large α small	LS
Bagging	— " —	—	—	LS
RF	— " —	m : nb. of features used randomly in each split	(m large)	LS
Boosting	— " —	B : nb. of trees d : splits of each tree (G_1) $\Rightarrow \eta$: shrinkage para.	B large d large η small	LS

Neural Network

model para.

$b^{(1)}, \dots, b^{(n+1)}$ bias
 $W^{(1)}, \dots, W^{(n+1)}$
weight matrices

trainig para.

width (nb. of neurons
in the layers)
depth (nb. of layers)
learning rate in GD
activation fct. (ReLU, sigmoid)
filter sizes
in CNN
early stopping (epochs)

High complexity

large width, depth
large nb. of epochs.
large nb. of FC layers

Fitters

LS, (stochastic) GD
↑
back-prop.

Classification methods: $x \in \mathbb{R}^p$, $y \in \mathcal{Y} = \{1, \dots, q\}$ classes

Method	Model par.	Tuning par.	High complex., if...	Fitting
LRNN	—	$h = 1, \dots, n$	h small	—
LPA	$\pi_{1, \dots, q}$ prior class prob.	—	(large p)	Gaussian ML (closed form)
QPA	$\mu_{1, \dots, q}$ means + Σ common cov. matrix + $\Sigma_{1, \dots, q} \in \mathbb{R}^{p \times p}$	—	— " —	— " —
log. reg. + lasso/ridge	$\vec{\beta} \in \mathbb{R}^{p+1}$ — " —	— $\lambda \geq 0$	(large p) λ small	Bernoulli ML (Multinomial)
Classification tree bagging / RF / boosting	as in regression			ML (deviance, cross-entropy) Gini index
MHC / SVC / SVM	$\vec{\beta} \in \mathbb{R}^{p+1}$ for MHC / SVC $\alpha_1, \dots, \alpha_n$ (SVM)	budget b cost $C = \frac{1}{b}$ + (SVM) $\gamma \geq 0$ in poly / Gaussian kernel	small b large γ	convex optimization

Neural networks
(CNN)

same as for regression
just the last fct.
is the soft max

ML (Bernoulli/
Multinomial)
deviance

Cross-validation

- gridSearchCV fits models for different tuning para. on a "grid"

e.g. boosting.

tree depth d

nb. of trees
 B

	1	2	3
1	•	•	•
2	•	•	•
3	•	•	•
4	•	•	•
⋮			
⋮			

each point is a model
fitted k times
 $\Rightarrow$ compute CV error
choose the best

best model in terms of CV
 $d=3$ $B=4$

$\Rightarrow$ fit model with $d=3, B=4$
again to whole training data